

ARAVIND GOVINDHASAMY

Full Stack Developer | IoT & Backend Engineer

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Summary

Full Stack Developer with hands-on experience across IoT platforms, backend services, AI-powered computer vision APIs, and responsive React dashboards. Currently building a cloud-based IoT platform at Semicon Media using FastAPI, MQTT, Supabase (PostgreSQL), and DigitalOcean, delivering real-time SMS/Email/WhatsApp alerting, GPS tracking (GeoLinker), and CV APIs like license plate, helmet, object, and face detection. Previously delivered enterprise ERP solutions at PSGR Krishnammal College (RFID Gate Software, GLPI for 1800+ assets, Journal Management) and Retail/CRM systems at 2CQR Automation. Strong in Python, C#, Node.js, SQL, REST API design, and full SDLC delivery.

Education

Master of Computer Application - First Class with Distinction, CGPA: 8.0

Sri Ramakrishna Mission Vidyalaya College of Arts and Science, Coimbatore (Bharathiyar University)
Aug 2022 — May 2024

Bachelor of Computer Science - First Class, CGPA: 7.46

Sri Ramakrishna Mission Vidyalaya College of Arts and Science, Coimbatore (Bharathiyar University)
Aug 2019 — May 2022

Higher Secondary Certificate - 74.9%

Govt Hr Sec School, Periyampatti (State Board of School Examinations, Tamil Nadu)

Work Experience

Semicon Media Pvt Ltd, Coimbatore

Dec 2025 — May 2026

Full Stack Developer

Building and deploying a cloud-based IoT platform (similar to Blynk / Arduino Cloud / ThingsBoard) enabling real-time device communication, monitoring, and alerting. Working across backend services, frontend dashboards, AI-powered computer vision APIs, real-time streaming, and cloud infrastructure.

- Built and deployed a cloud-based IoT platform enabling real-time device communication and monitoring using MQTT protocols.
- Designed and developed backend services using FastAPI, handling asynchronous processing of IoT data, messaging workflows, and API integrations.
- Implemented real-time notification systems supporting SMS, Email, and WhatsApp, enabling global alerting from IoT devices with template-based messaging.
- Developed GeoLinker, a real-time GPS tracking system with geofencing, route history, and live map visualization using React.
- Built and integrated multiple AI-powered computer vision APIs including license plate recognition, object detection, helmet detection, and face detection for security and analytics use cases.
- Implemented WebRTC-based real-time communication features for low-latency data/video streaming in IoT and monitoring applications.
- Designed and worked with MCP (Modular/Message Control Platform) architecture, enabling scalable communication between microservices and IoT components.
- Used Supabase (PostgreSQL) for database management, authentication, and real-time data handling.
- Deployed and maintained applications on DigitalOcean (1-core VPS), managing backend services, API hosting, and uptime reliability.
- Integrated Sentry for real-time error tracking, monitoring, and production debugging.
- Designed scalable REST APIs and optimized database queries for low-latency responses in real-time applications.
- Built responsive frontend dashboards using React for live device monitoring, alerts, and analytics visualization.
- Managed version control and collaboration using GitHub, maintaining clean code practices and deployment workflows.

PSGR Krishnammal College for Women, Coimbatore, Tamil Nadu

Oct 2024 — Nov 2025

ERP Assistant Programmer

Working in the Systems Department, responsible for designing, developing, customizing, and maintaining enterprise software solutions to support insti-

Jun 2016 — May 2017

Secondary School Certificate - 86%

Govt Hr Sec School, Periyampatti
(State Board of School
Examinations, Tamil Nadu)

Jun 2014 — May 2015

Programming Languages

Python, C#, Node.js, JavaScript,
TypeScript

Backend Frameworks

FastAPI, .NET, .NET Core, Express.js

Frontend

React, HTML5, CSS3, Tailwind CSS,
Bootstrap

Databases

PostgreSQL, Supabase, MySQL,
MariaDB, MS SQL Server

Cloud & DevOps

DigitalOcean, Supabase, Sentry, GitHub
Actions, CI/CD

IoT & Hardware

MQTT, Raspberry Pi, IoT RFID, RFID
Reader Integration, GPS / Geofencing

AI / Computer Vision

License Plate Recognition, Object
Detection, Helmet Detection, Face
Detection, Gemini AI

Messaging & Notifications

SMS, Email, WhatsApp API, Template-
based Messaging

Web Technologies

REST API, JSON, WebSockets,
WebRTC, MCP Architecture

Environments & Tools

GitHub, Visual Studio, VS Code,
Postman, Jupyter Notebook, WSL,
PowerShell

Operating Systems

Windows, Linux

Productivity Tools

Microsoft Office, Canva, ChatGPT

Soft Skills

Creativity, Problem-solving,
Communication, Time management,

tutional operations across full SDLC phases including requirement gathering, design, development, testing, and deployment.

- Developed RFID-based Gate Software Ecosystem with real-time monitoring and camera integration, serving 5000+ users.
- Designed SQL-driven reporting systems for user movement analysis and trend detection.
- Built ID Card Tagging module supporting HF, UHF, and QR cards with duplicate detection.
- Customized GLPI for IT asset tracking across 1800+ systems with real-time inventory updates.
- Implemented Journal Management System with full editorial workflow and role-based access supporting complete peer review.
- Conducted R&D and requirement analysis for Blood Bank Management System, designing modular architecture for phased development.
- Created analytics dashboards and automated daily report generation with email delivery.
- Collaborated with stakeholders to ensure alignment of system features with operational requirements.

2CQR Automation Private Limited, Coimbatore

May 2024 — Oct 2024

Junior Software Developer

Developed and maintained Library and Jewellery ERP Automation systems integrated with RFID, along with Retail, Customer Feedback, and CRM applications using Node.js APIs and C#. Worked on backend development and integration of external hardware (RFID readers) for end-to-end inventory tracking and operations.

- Built Library and Jewellery ERP Automation modules integrated with RFID for real-time inventory and transaction tracking.
- Delivered multiple high-priority projects in under 3 months, demonstrating quick adaptability to new technologies.
- Successfully integrated multiple APIs and external systems into a cohesive, well-functioning product.
- Quickly learned Node.js and efficiently handled both C# and Node.js development simultaneously.
- Identified customer-side issues promptly, providing effective solutions and showcasing strong debugging and troubleshooting skills.
- Collaborated with cross-functional teams to deliver solutions that met client requirements within tight deadlines.

2CQR Automation Private Limited, Coimbatore

Jan 2024 — May 2024

.NET Developer Intern

Developed desktop applications in C# for library automation systems, integrating RFID readers for seamless operation. Worked on backend tasks including API development, database management, and hardware integration.

- Successfully delivered a C# desktop application for automating library systems using RFID technology.
- Contributed to backend solutions that significantly improved operational efficiency for clients.
- Implemented GitHub workflow automation in the company, enhancing team collaboration and development efficiency.

Languages

English, Tamil

Certifications

- NCC - B & C Certificates (NCC India, 2022)
- ChatGPT for Everyone (GUVI, Dec 2023)
- Programming Using Python (GUVI, Dec 2023)
- Game Development using PyGame (GUVI, March 2024)

- Gained hands-on experience in .NET Core, C#, Python, and SQL, contributing to real-world solutions.
- Learned how to effectively integrate hardware (RFID readers) with software applications for practical automation.

Projects

Circuit Digest Cloud - IoT Platform (Semicon Media Project) Dec 2025 — May 2026

<https://www.circuitdigest.cloud/>

A cloud-based IoT platform (similar to Blynk / Arduino Cloud / ThingsBoard) enabling real-time device communication, monitoring, alerting, GPS tracking, and AI-powered computer vision APIs.

- Technologies Used: FastAPI, MQTT, Supabase (PostgreSQL), React, WebRTC, DigitalOcean, Sentry
- Built MQTT-based real-time device communication and monitoring backend.
- Implemented SMS, Email, and WhatsApp alerting with template-based messaging.
- Developed GeoLinker - real-time GPS tracking with geofencing and route history.
- Integrated AI-powered CV APIs for license plate, object, helmet, and face detection.
- Implemented WebRTC for low-latency data/video streaming.
- Designed MCP architecture for scalable microservice communication.

Journal Management System - AAR PSGRKCW (PSGRKCW Project) Oct 2024 — Nov 2025

<https://journalpaper.psgrkcw.ac.in/index.php/aar-psgrkcw>

Implemented a complete Journal Management System for PSGR Krishnammal College of Women (Annals of Arts Research) supporting the full editorial workflow - submission, peer review, editorial decisions, and publication with role-based access.

- Technologies Used: OJS (Open Journal Systems), PHP, MySQL, Linux Server
- Deployed and customized the journal platform end-to-end.
- Configured editorial workflow with author/reviewer/editor roles.
- Set up secure hosting and maintained uptime.
- Trained editorial staff on managing submissions and reviews.

GLPI - IT Asset Tracking System (GRG Education / PSGRKCW Project) Oct 2024 — Nov 2025

<http://glpi.grgeducation.com/>

Customized and deployed GLPI (open-source IT asset management) to automate inventory tracking across 1800+ systems with real-time updates, helpdesk ticketing, and lifecycle management for the entire GRG Education institution.

- Technologies Used: GLPI, PHP, MySQL, Linux, FusionInventory Agent
- Automated inventory tracking for 1800+ IT assets with real-time updates.
- Configured FusionInventory agents for auto-discovery across departments.
- Built helpdesk ticketing workflow integrated with asset records.

- Customized GLPI plugins and dashboards for institutional reporting.

Bude Global Neuro-Chain - Innovation Network

Jun 2025 —

Visualization Platform (BUDE Global Enterprise Project)

May 2026

<https://invent.budeglobal.in/>

Interactive visualization of human innovation as a non-linear network, showing how technologies build upon each other from fire to AGI. Powered by Bude Global.

- Technologies Used: React, JavaScript, HTML5, CSS3, Graph Visualization
- Built an interactive non-linear graph of innovations spanning from fire to AGI.
- Designed responsive React UI for exploring technology evolution and dependencies.
- Implemented dynamic node-link visualization for storytelling around human innovation.
- GitHub: <https://github.com/BUDEGlobalEnterprise/bude-global-neuro-chain-react>

BUDE Global - Tech Presentations

Jun 2025 — May 2026

<https://github.com/BUDEGlobalEnterprise/bude-global-tech-presentations>

A curated collection of technical presentations and slide decks covering enterprise tech topics, architecture patterns, and product walkthroughs maintained for the BUDE Global Enterprise team.

- Technologies Used: Markdown, Slide Frameworks, GitHub Pages
- Authored and organized technical presentations for internal and external use.
- Maintained version-controlled slide decks with clean structure.
- Contributed to documentation and knowledge sharing across the enterprise.

Bitcoin Mining Setup using Raspberry Pi and Kali Linux (UG Project)

Jan 2022 — Apr 2022

<https://github.com/aravind-govindhasamy/BifidCipher-Web-Tool>

Developed a low-cost Bitcoin mining system using Raspberry Pi and open-source software. The project explored the feasibility of mining cryptocurrencies with energy-efficient hardware.

- Technologies Used: Raspberry Pi, Kali Linux, Python
- Configured the Raspberry Pi for optimal performance.
- Implemented the Bitcoin mining software on Kali Linux.
- Analyzed the mining performance and adjusted configurations to maximize efficiency.

IoT-Based Smart Library System using RFID (2CQR Project)

Jan 2023 — May 2023

<https://github.com/aravind-govindhasamy/IoT-Based-Smart-Library-System-using-RFID>

Developed a smart library system using RFID and IoT technology to automate book check-in and check-out processes. Allows real-time monitoring of book inventory and patron transactions, integrated with SQL Server for backend data storage.

- Technologies Used: C#, ASP.NET, SQL Server, RFID, IoT
- Designed the user interface using C# and ASP.NET. <https://atsresume.vercel.app/>

- Integrated RFID readers for real-time tracking of book movements.
- Created SQL Server databases to store book and patron information.
- Optimized the backend to improve transaction speed and accuracy.

Library Fine Collection Kiosk Software Using RFID and Razor Payment Gateway (2CQR Project)

Aug 2023 —
Dec 2023

<https://github.com/aravind-govindhasamy/Library-Fine-Collection-Kiosk>

Developed a desktop application to automate library fine collection using RFID technology. Integrated Razor payment gateway for seamless payment processing. The system provides a user-friendly kiosk interface for patrons to check and pay fines.

- Technologies Used: C#, .NET, RFID, Razor Payment Gateway, MSSQL
- Designed and developed the desktop form application using C# and .NET.
- Integrated RFID to track books and user data for fine calculation.
- Implemented the Razor payment gateway for secure, real-time fine payments.
- Managed data storage and retrieval with MSSQL for fine transactions and user details.

Stock Control Inventory Management Application Using RFID (2CQR Project)

Jan 2024 —
May 2024

<https://github.com/aravind-govindhasamy/Stock-Control-Inventory-Management>

Contributed to the development of an inventory management application using RFID to track and manage stock levels in real-time. Worked on backend API development and stored procedures for handling inventory data and reporting.

- Technologies Used: C#, ASP.NET, RFID, MSSQL, REST API
- Developed and maintained backend REST APIs for inventory management operations.
- Worked on MSSQL stored procedures for efficient data retrieval and manipulation.
- Integrated RFID technology to automate stock control and inventory tracking.
- Optimized database queries to enhance performance and reduce response times.

Bulk Image Downloader from CSV Using Streamlit

Jun 2024 — Aug 2024

<https://github.com/GamersNearMe/Bulk-Image-Downloader-from-CSV-using-streamlit>

A Streamlit app that allows users to upload CSVs, select columns containing image URLs, and bulk download images efficiently.

- Technologies Used: Python, Streamlit, CSV
- Developed the Streamlit app for CSV uploads and image processing.
- Implemented functionality to parse URLs and download images efficiently.
- Designed a user-friendly interface for selecting and managing downloads.
- Included logging features to track download progress and issues.

Python Licensing System

Sep 2024 — Nov 2024

<https://github.com/aravind-govindhasamy/PythonLic>

A licensing system in Python using cryptography for encryption and MAC address validation, designed for secure license generation and management.

- Technologies Used: Python, Cryptography
- Implemented AES encryption for secure license generation and validation.
- Developed MAC address validation for license tracking.
- Created a user interface for managing licenses and users.

BulkDBBackup NodeJS MySQL

Jun 2024 — Sep 2024

https://github.com/aravind-govindhasamy/BulkDBBackup_NodeJS_MySQL

MySQL backup script that backs up all user databases from a MySQL server with scheduling and detailed logging.

- Technologies Used: Node.js, MySQL
- Automated the backup process, reducing manual intervention.
- Implemented error handling to ensure backup integrity and reliability.
- Developed a scheduling feature to run backups at specified intervals.
- Created a detailed log system to track backup activities and outcomes.

AES Encryption and Decryption Using CryptoJS

Mar 2024 — Apr 2024

<https://github.com/aravind-govindhasamy/AES-Encryption-Decryption>

Demonstrates AES encryption and decryption using CryptoJS in HTML/JavaScript with a user-friendly interface.

- Technologies Used: JavaScript, HTML, CryptoJS
- Implemented secure data transmission using AES encryption standards.
- Designed a user-friendly interface for easy input and output handling.
- Utilized best practices for cryptographic functions to enhance security.
- Provided detailed documentation for users on encryption/decryption processes.

PDF Image Extractor Using Gemini 1.5 AI

Oct 2024 — Dec 2024

<https://github.com/aravind-govindhasamy/Invoice-Extractor-LLM-APP>

Streamlit app that extracts images from each page of uploaded PDF files using PyMuPDF for smooth PDF processing.

- Technologies Used: Python, Streamlit, PyMuPDF, Gemini 1.5 AI
- Extracted images from each page of uploaded PDF files.
- Displayed extracted images seamlessly in the app.
- Integrated Gemini 1.5 AI for intelligent invoice/document understanding.